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Title: Non- r -partite graphs without complete split subgraphs.

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Review text:

Let $T_{n,r}$ denote the r -partite Turán graph on n vertices, and let $\rho(G)$ denote the spectral radius of the graph G . The spectral version of Turán's theorem due to Nosal [*Eigenvalues of graphs*, Master's thesis, Univ. of Calgary (1970)] and Nikiforov [Linear Algebra Appl. **427** (2007), no. 2–3, 183–189; MR2351351] says that $\rho(G) \leq \rho(T_{n,r})$ for any K_{r+1} -free graph G on n vertices, with equality if and only if $G \cong T_{n,r}$. Several variations on this theme have been considered recently in the literature under the umbrella of spectral Turán-type problems.

Given a graph H , let $\text{EX}_{r+1}(n, H)$ (resp., $\text{SPEX}_{r+1}(n, H)$) denote the collection of extremal (resp., spectral extremal) graphs for the class of non- r -partite H -free graphs on n vertices, up to isomorphism. Let $Y_r(n)$ be the graph obtained from the Turán graph $T_{n-1,r}$ by removing an edge u_1u_2 between two of the smallest parts and adding a vertex u_0 that is adjacent to u_1, u_2 , and all the vertices in the rest of the parts. It is known from the work of Brouwer [Afdeling Zuivere Wiskunde [Department of Pure Mathematics], 152. Mathematisch Centrum, Amsterdam, 1981; MR0611510] that $Y_r(n) \in \text{EX}_{r+1}(n, K_{r+1})$, and from the works of Lin–Ning–Wu [Combin. Probab. Comput. **30** (2021), no. 2, 258–270; MR4225787] and Li–Peng [SIAM J. Discrete Math. **37** (2023), no. 4, 2462–2485; MR4656787] that $\text{SPEX}_{r+1}(n, K_{r+1}) = \{Y_r(n)\}$.

In this paper, the authors extend these results by showing that, for $r \geq 3$, $q \geq 1$ and n sufficiently large, $Y_r(n) \in \text{EX}_{r+1}(n, B_{r,q})$ and $\text{SPEX}_{r+1}(n, B_{r,q}) = \{Y_r(n)\}$, where $B_{r,q}$ is the *generalized book graph* (or, *complete split graph*) defined as the join of K_r and qK_1 . Note that $B_{r,1} \cong K_{r+1}$. They also show

that $\text{EX}_{r+1}(n, K_{r+1}) \subseteq \text{EX}_{r+1}(n, B_{r,q})$ for $r \geq 3$, $q \geq 1$ and n sufficiently large. Since K_{r+1} is a subgraph of $B_{r,q}$, these results imply the result of Li–Peng (loc. cit.) for sufficiently large n .

Some minor typos: the statement of Theorem 1.4 is missing the hypothesis that G is non- r -partite, and equation (9) should read

$$\rho(G_\ell) - \rho(G) \geq \mathbf{x}^\top (A(G_\ell) - A(G)) \mathbf{x} \geq \sum_{i \in [\ell-1]} 2x_{u_i} (x_{v_i} - x_{u'}) \geq 0.$$